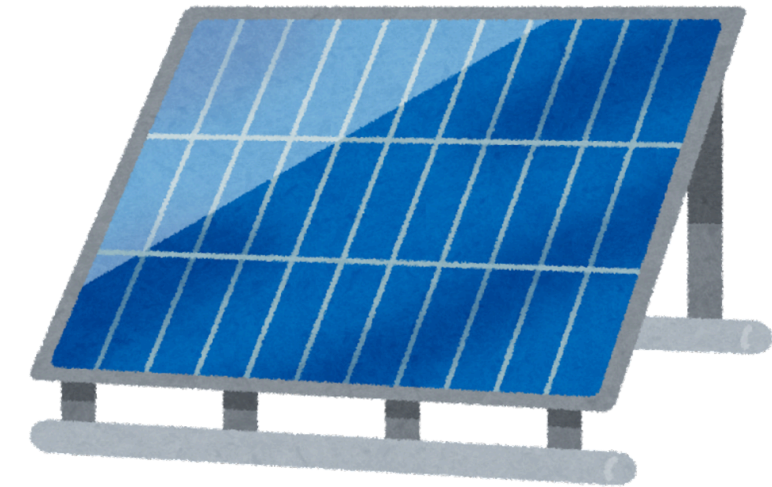


UST Campus Solar Solutions



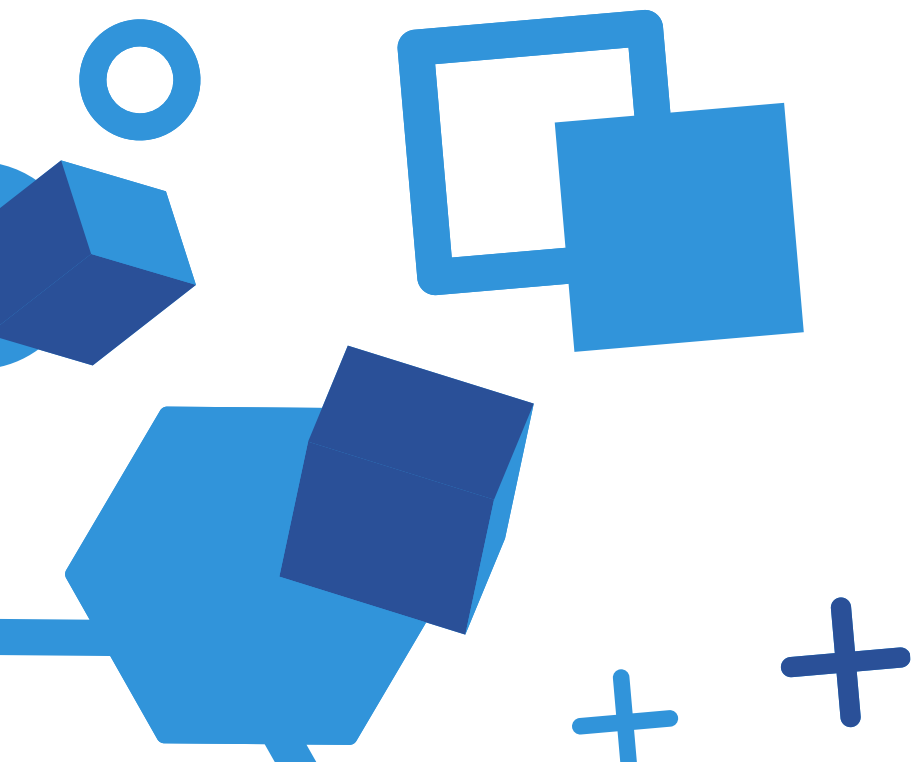
Optimizing Solar PV Performance

Presented By: **Gen Solar G**
Camen, Tiffany, Joyce, Joanne



Primary Goal:

To provide a comprehensive understanding of the status of the HKUST solar panel system, thereby deriving feasible action recommendations for optimization.



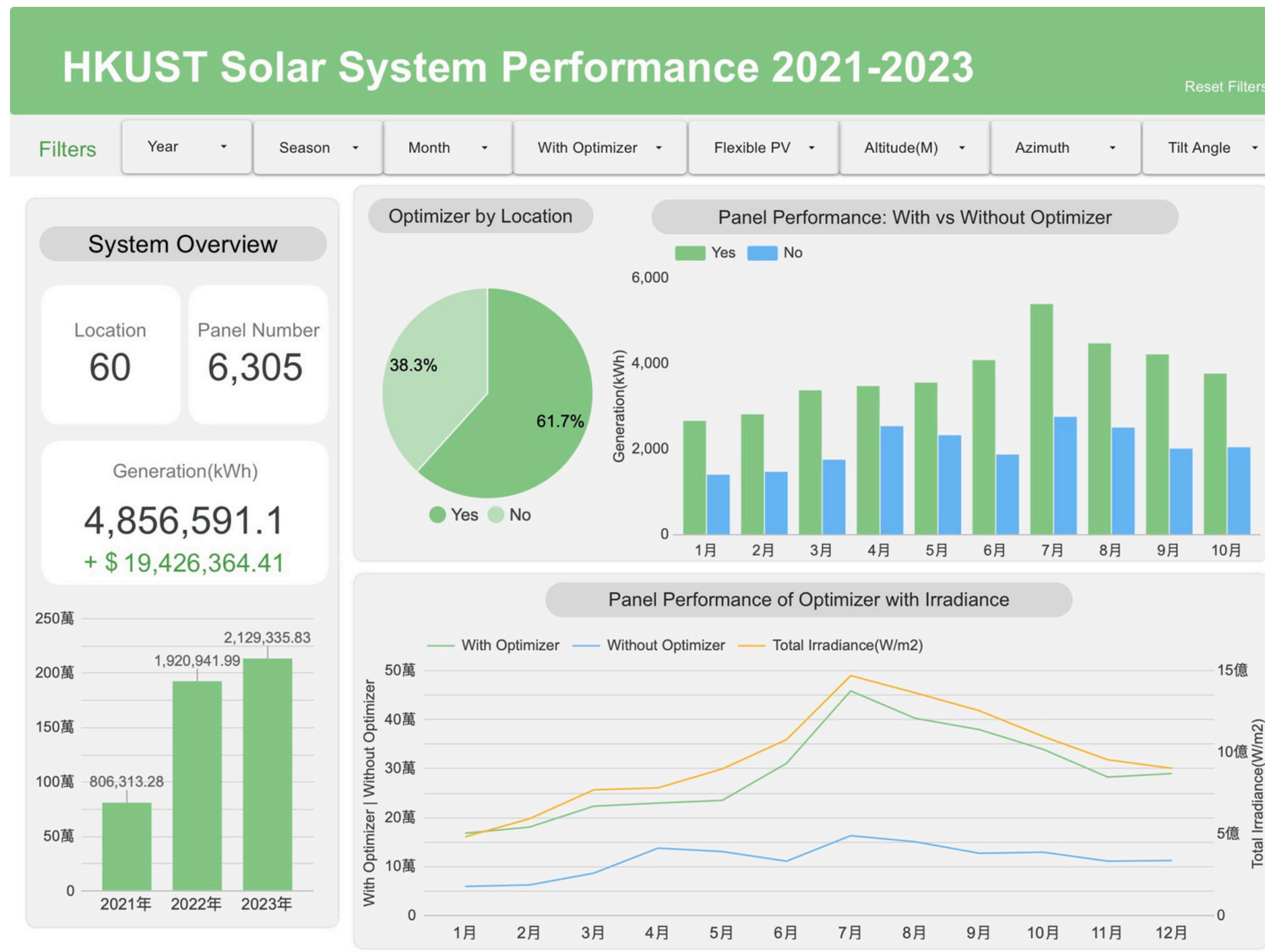
Specific Objectives:

- **Visual Insight :**
 - To consolidate multi-source data into a Diagnostic Dashboard
- **Event-based detection:**
 - To identify total down day and extreme performance volatility.
- **Environmental Correlation:**
 - To distinguish natural fluctuations from actual mechanical underperformance.
- **Cross-location Benchmarking:**
 - To prioritize high-risk locations for maintenance by benchmarking monthly performance (Z-score) against regional peers.



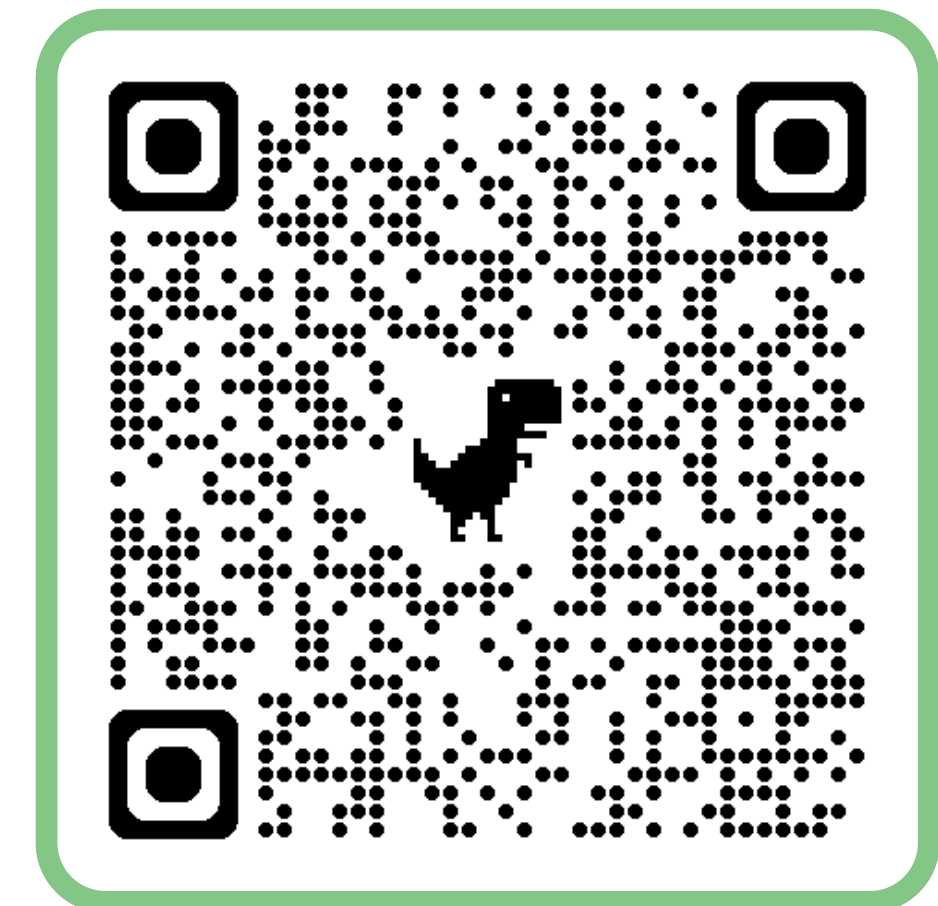
Dashboard

Understanding Performance of Solar System

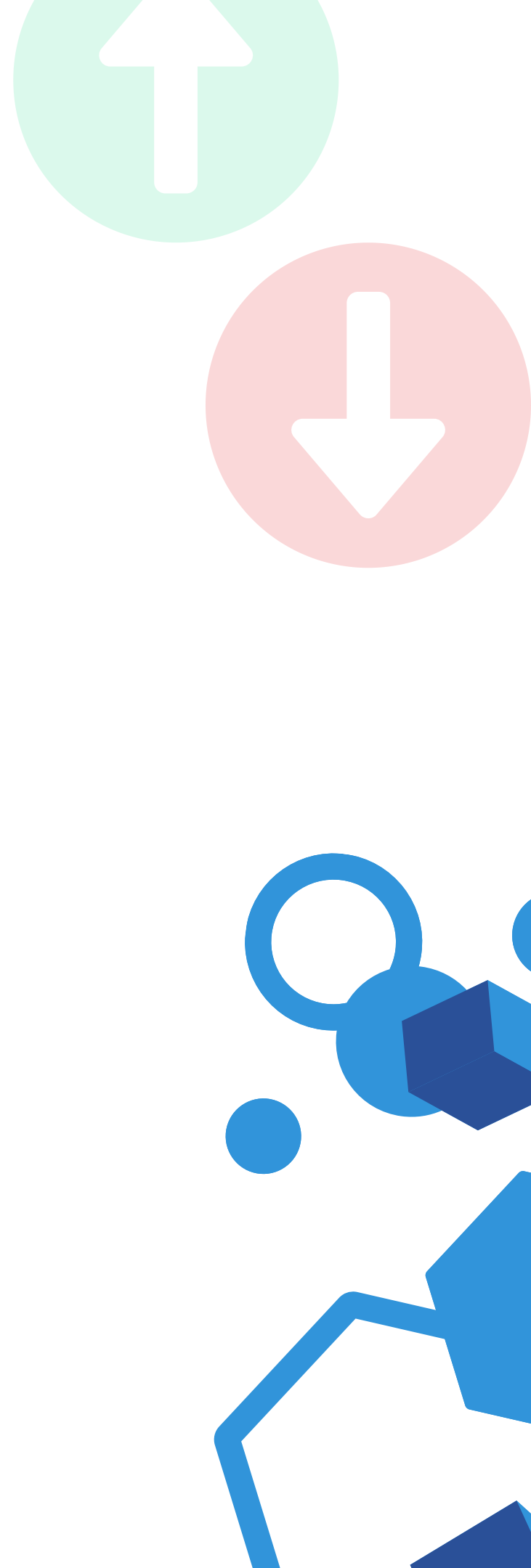
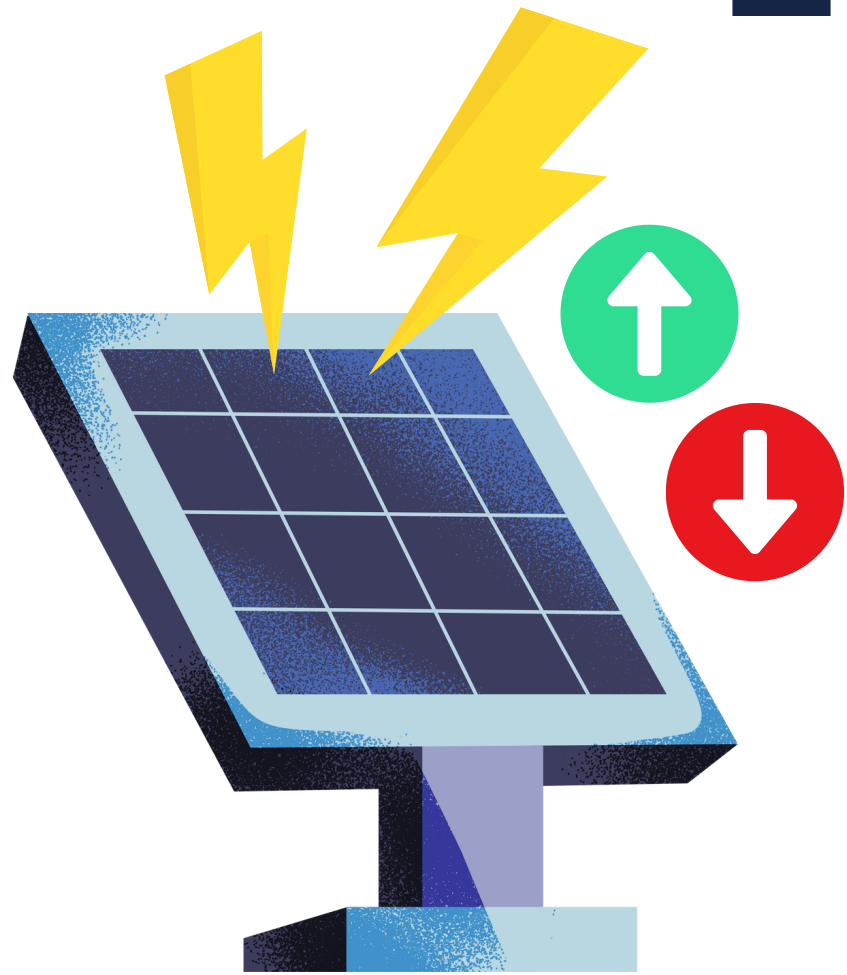


Looker Studio Link:

<https://lookerstudio.google.com/reporting/e704f941-d817-4bda-a830-b84b5f562005>



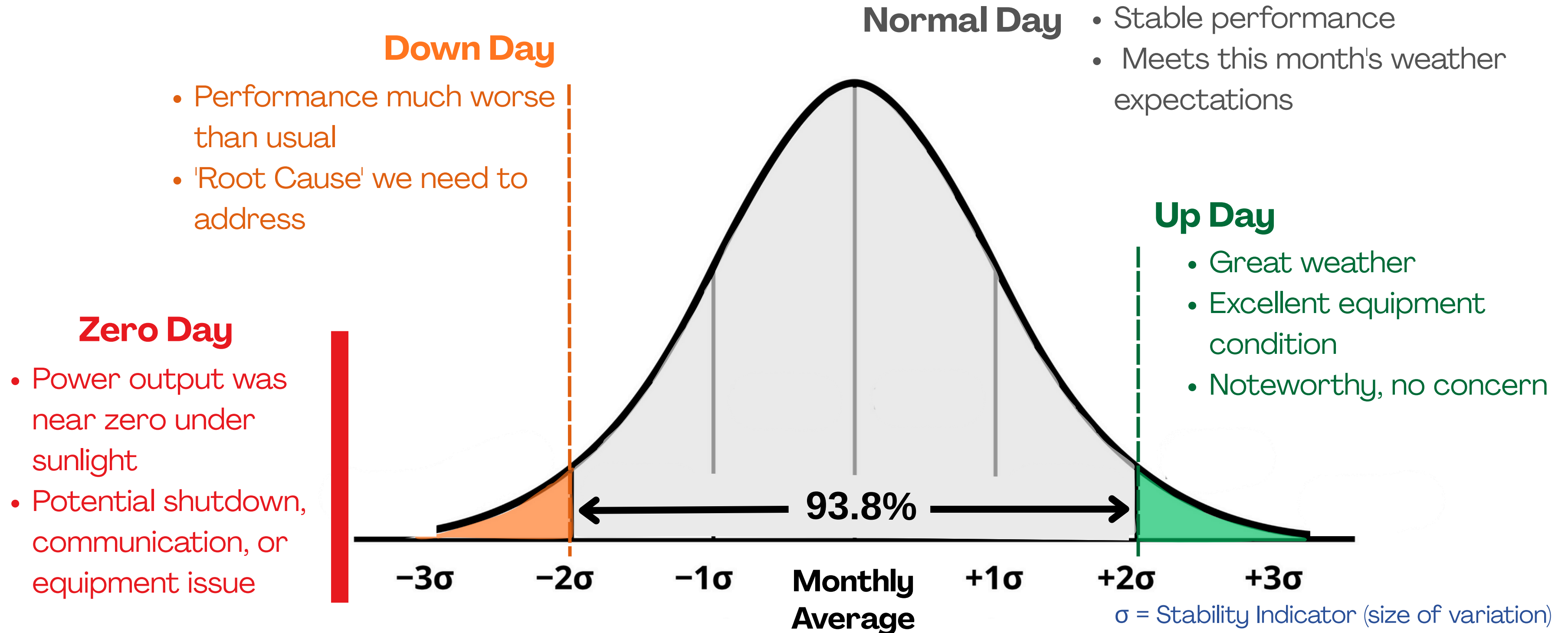
Event-Based Detection



How Daily Performance Is Interpreted?

Spotting True Anomalies with 2σ

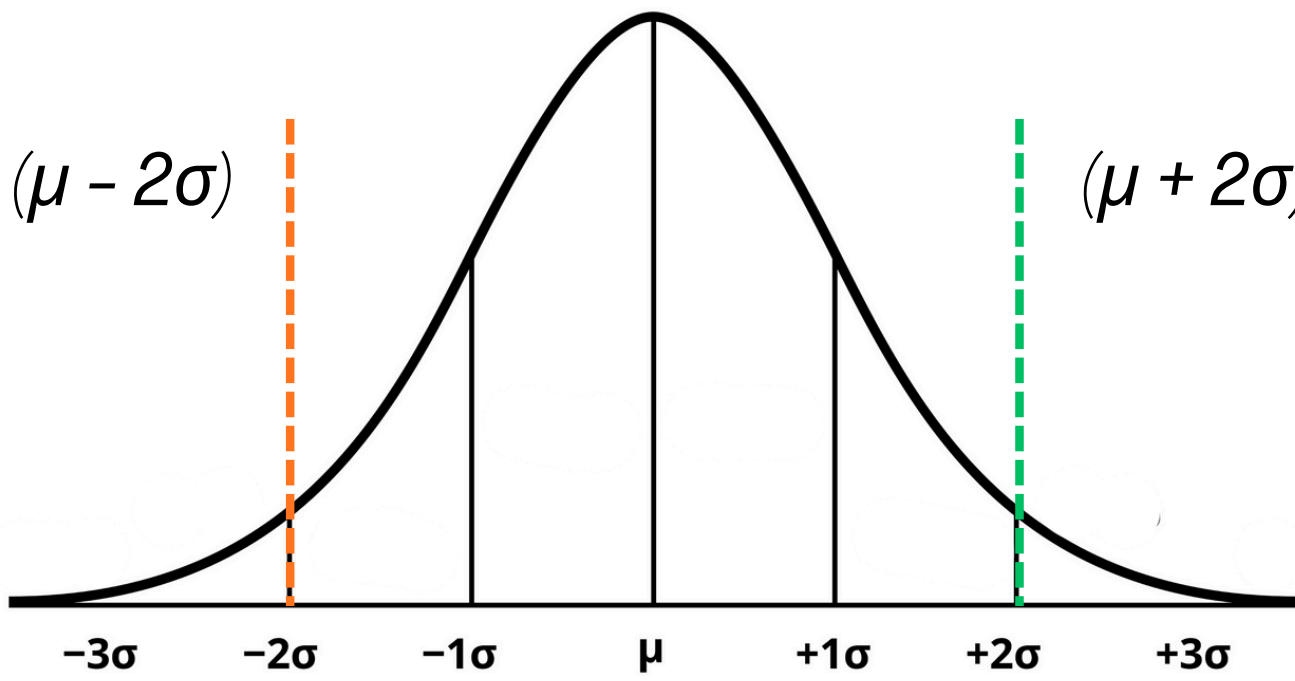
Monthly Average $\pm 2\sigma$



How Does 2σ Work?

Auto-adjusting for Seasons to Pinpoint Real Anomalies

Monthly Average $(\mu) \pm 2\sigma$



SQ 10

Date	Monthly Average	Lower Bound ($\mu - 2\sigma$)	Energy (kWh)	Upper Bound ($\mu + 2\sigma$)	Label
13/12/2022	62.84	13.63	> 13.45	112.06	Down Day
2/8/2023	79.42	15.44	> 148.00	143.39	Up Day

Why Solar Days Aren't Symmetric?

The Hidden Left Skew

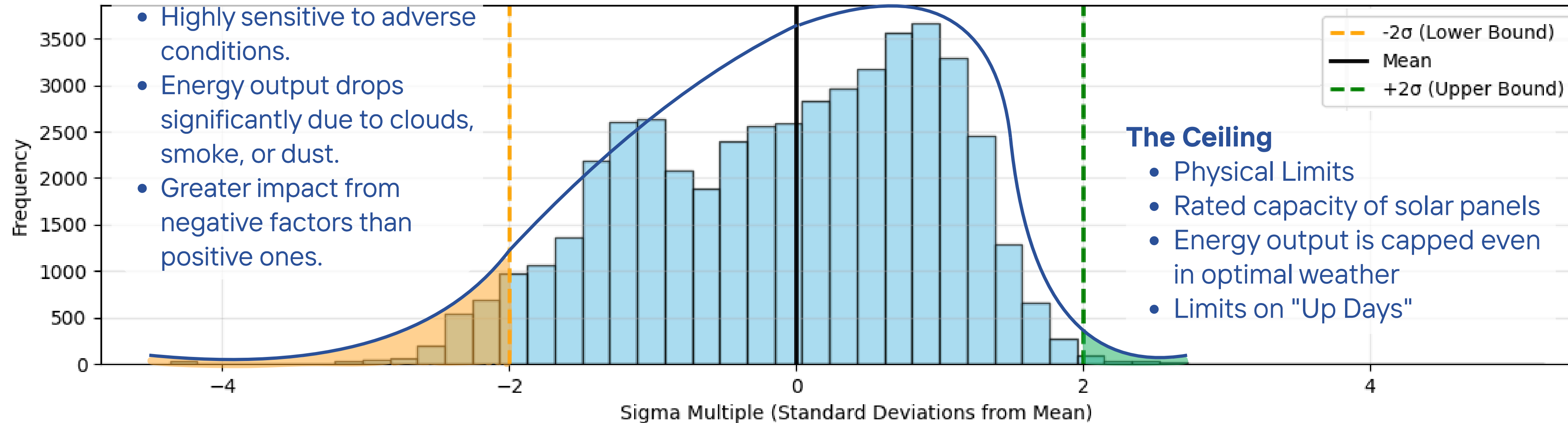
Overall Sigma Distribution for All 60 locations

The Floor

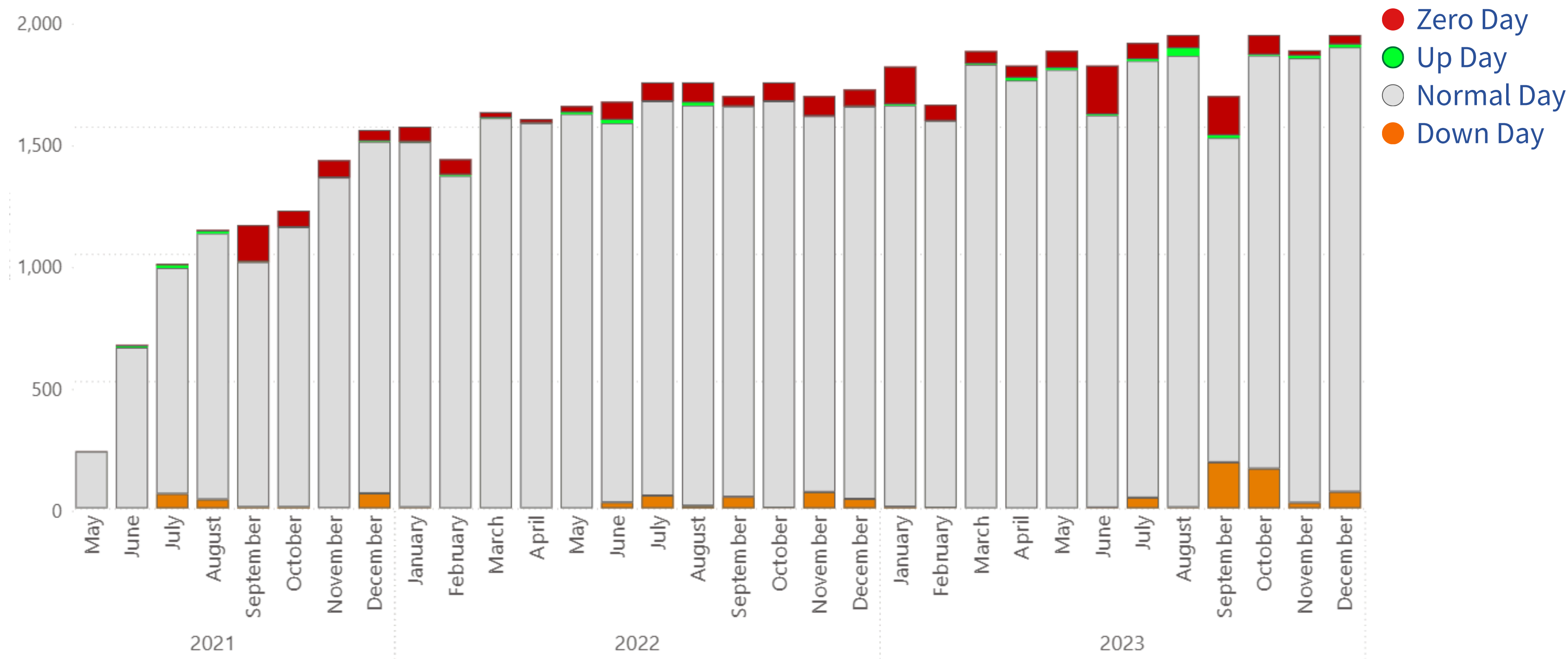
- Highly sensitive to adverse conditions.
- Energy output drops significantly due to clouds, smoke, or dust.
- Greater impact from negative factors than positive ones.

The Ceiling

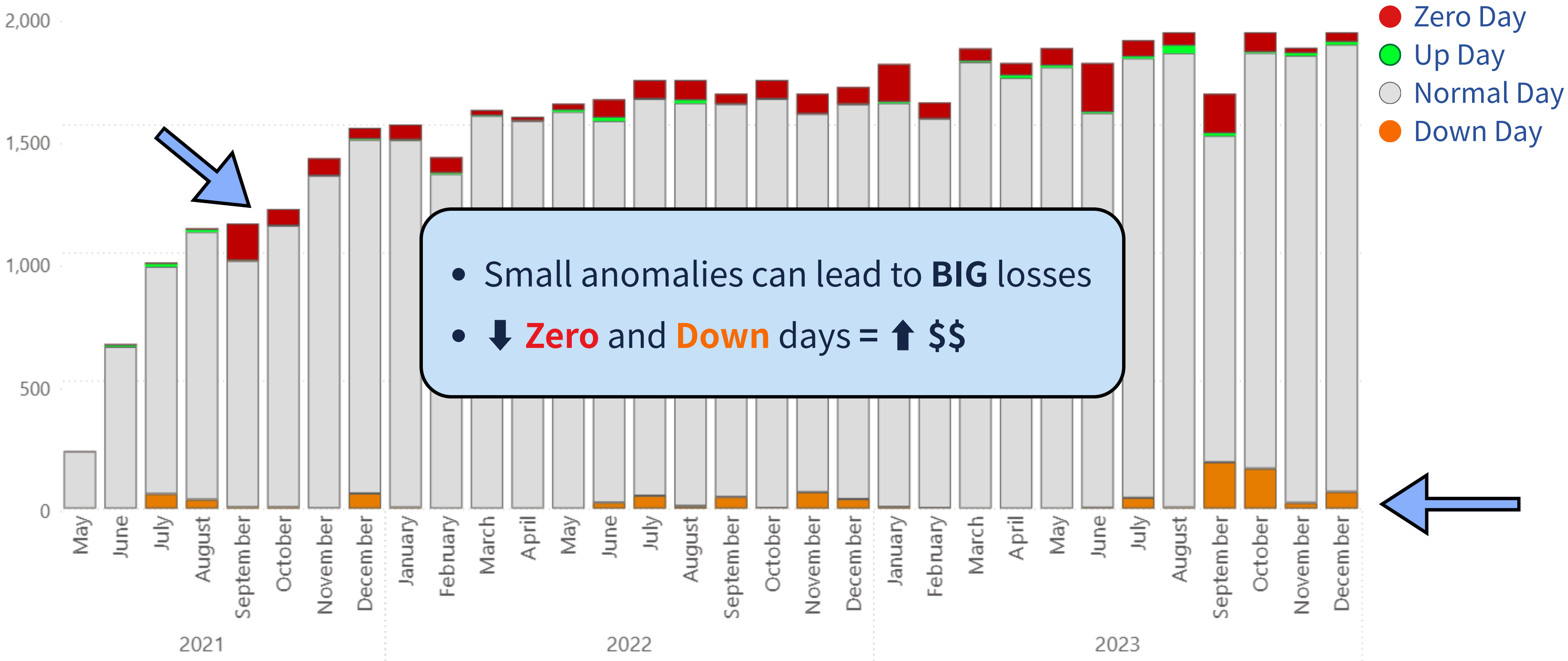
- Physical Limits
- Rated capacity of solar panels
- Energy output is capped even in optimal weather
- Limits on "Up Days"



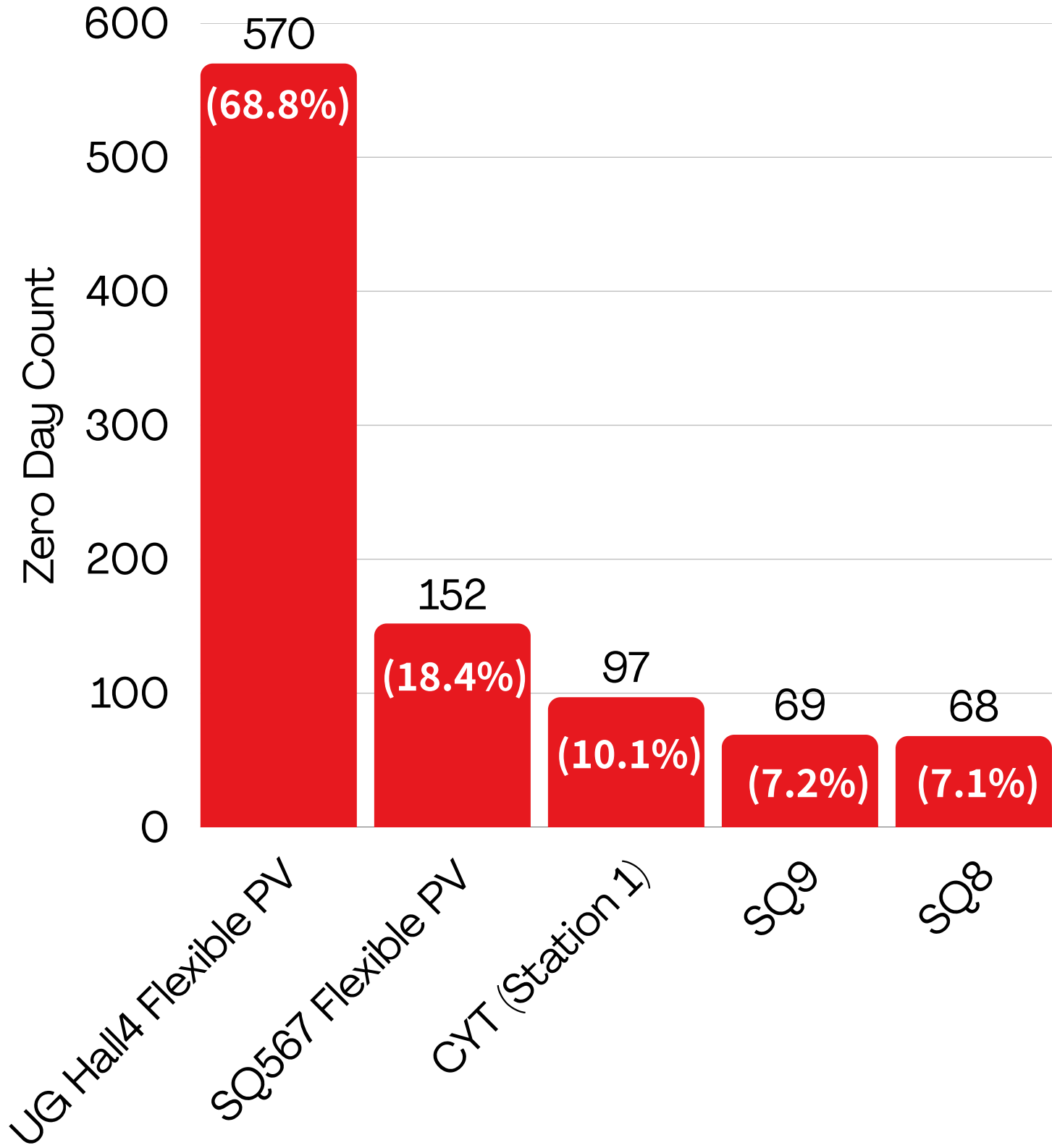
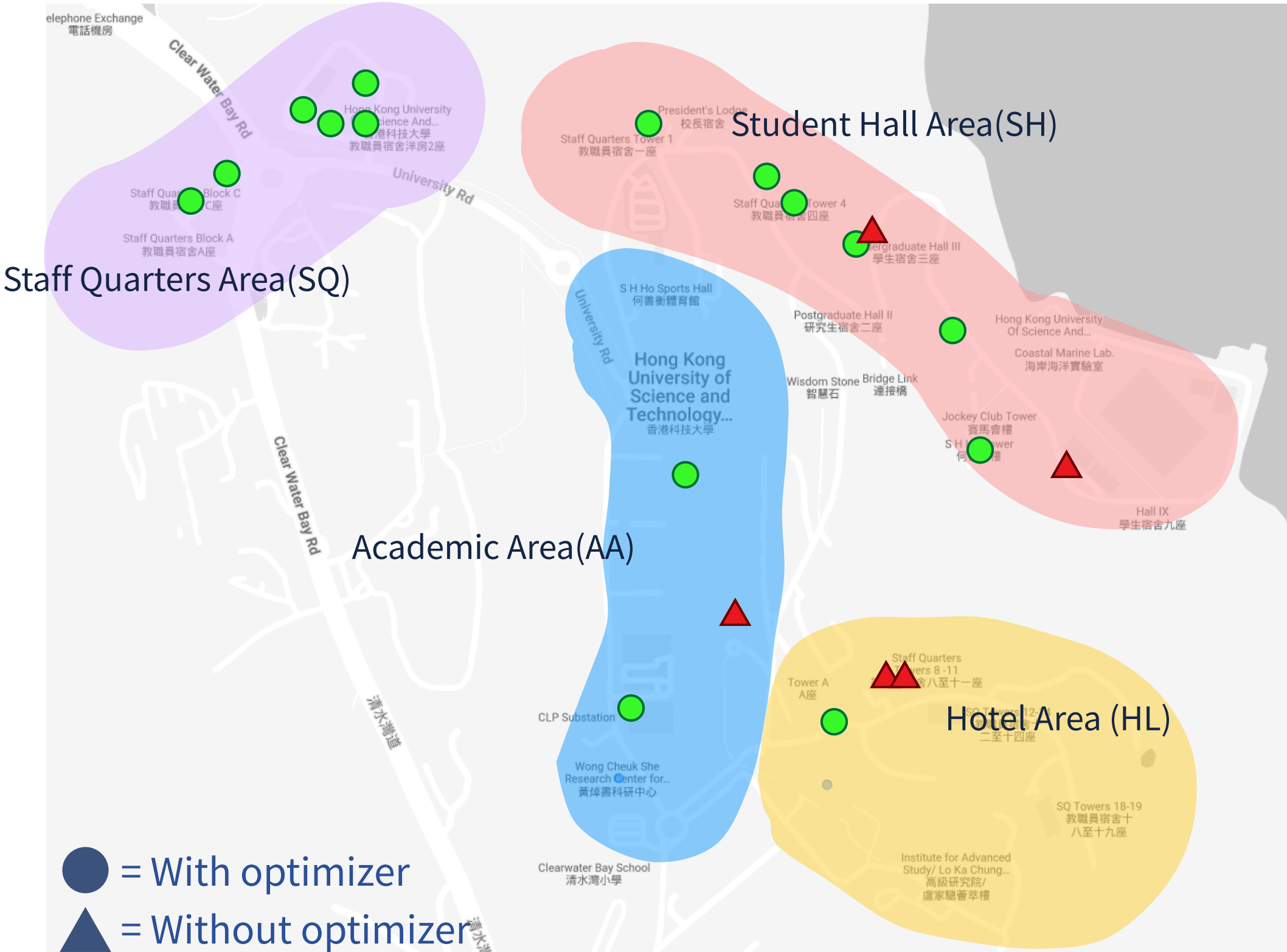
How Has Day Type Composition Evolved Over Time? (2021–2023)



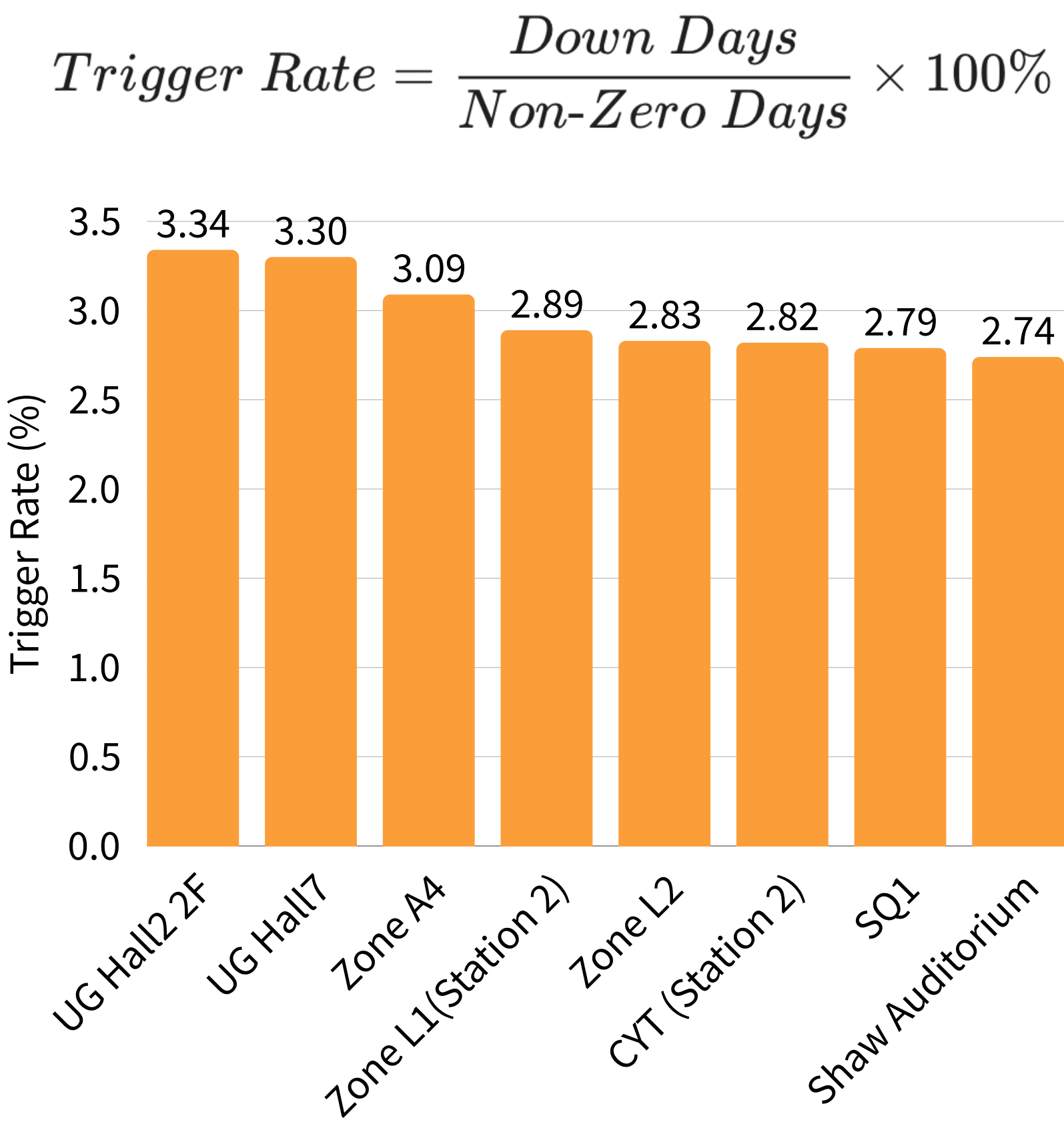
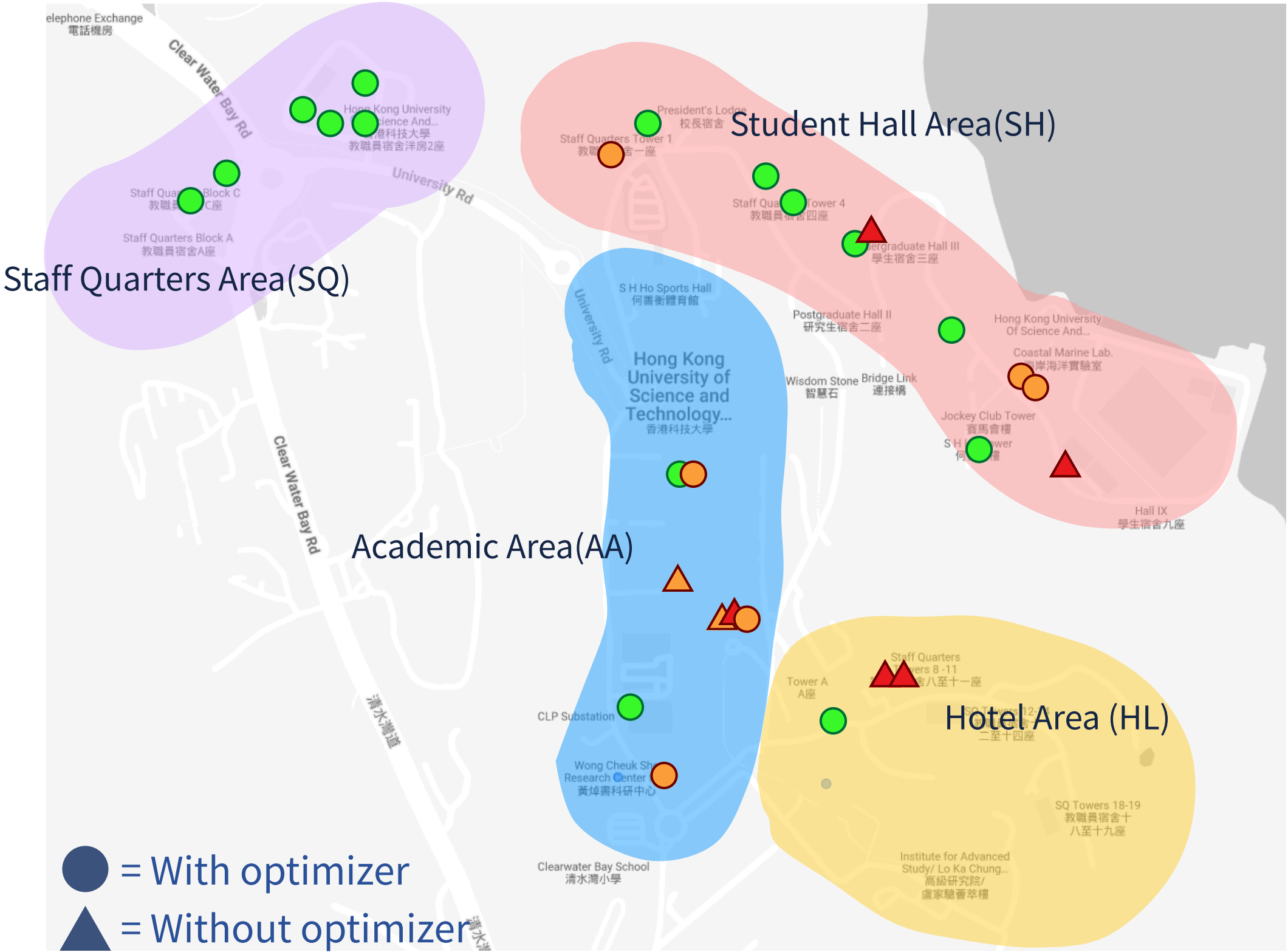
How Has Day Type Composition Evolved Over Time? (2021–2023)

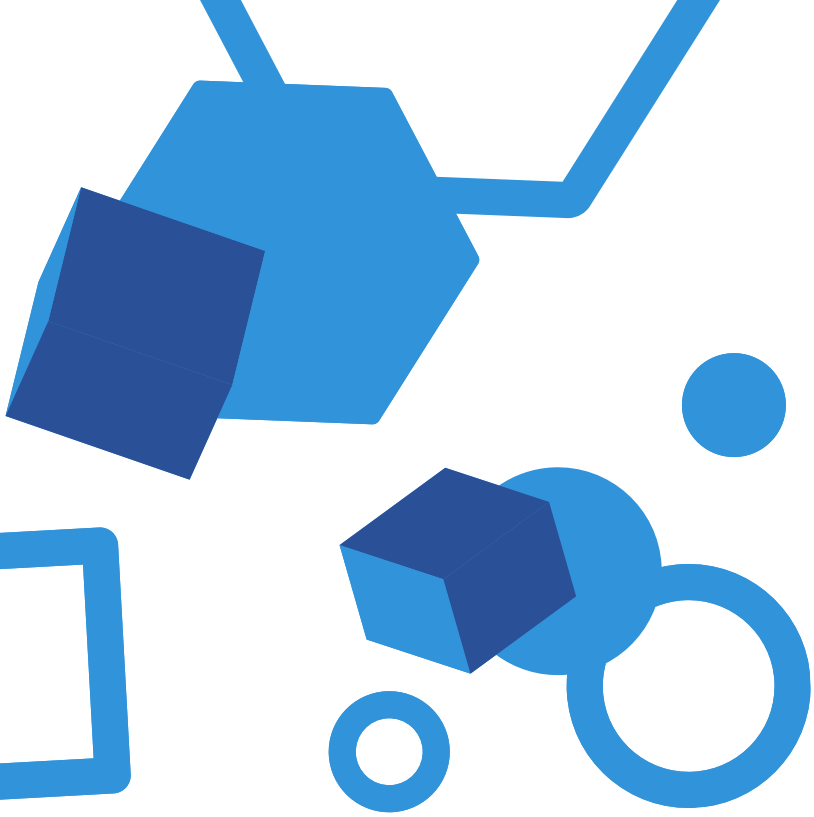


Zero Day Hotspots: Where Do Outages Concentrate?

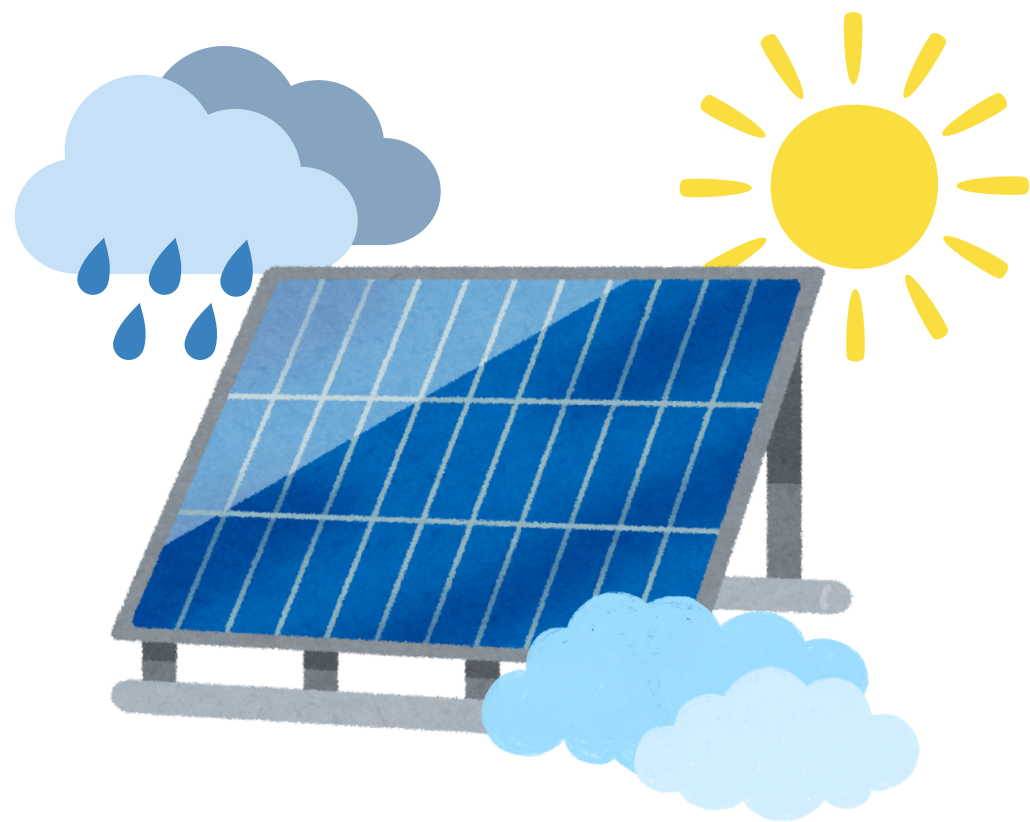


Where Are Down Days Most Concentrated?





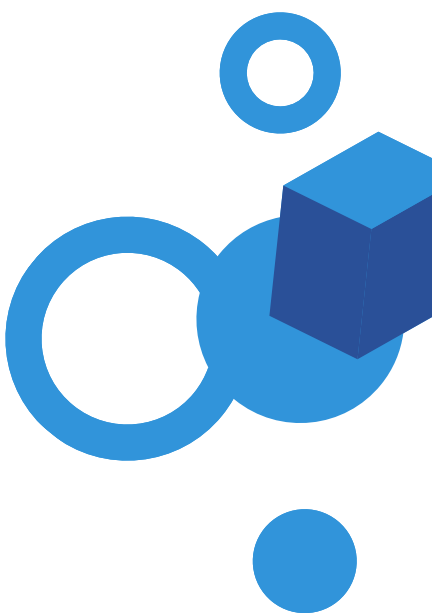
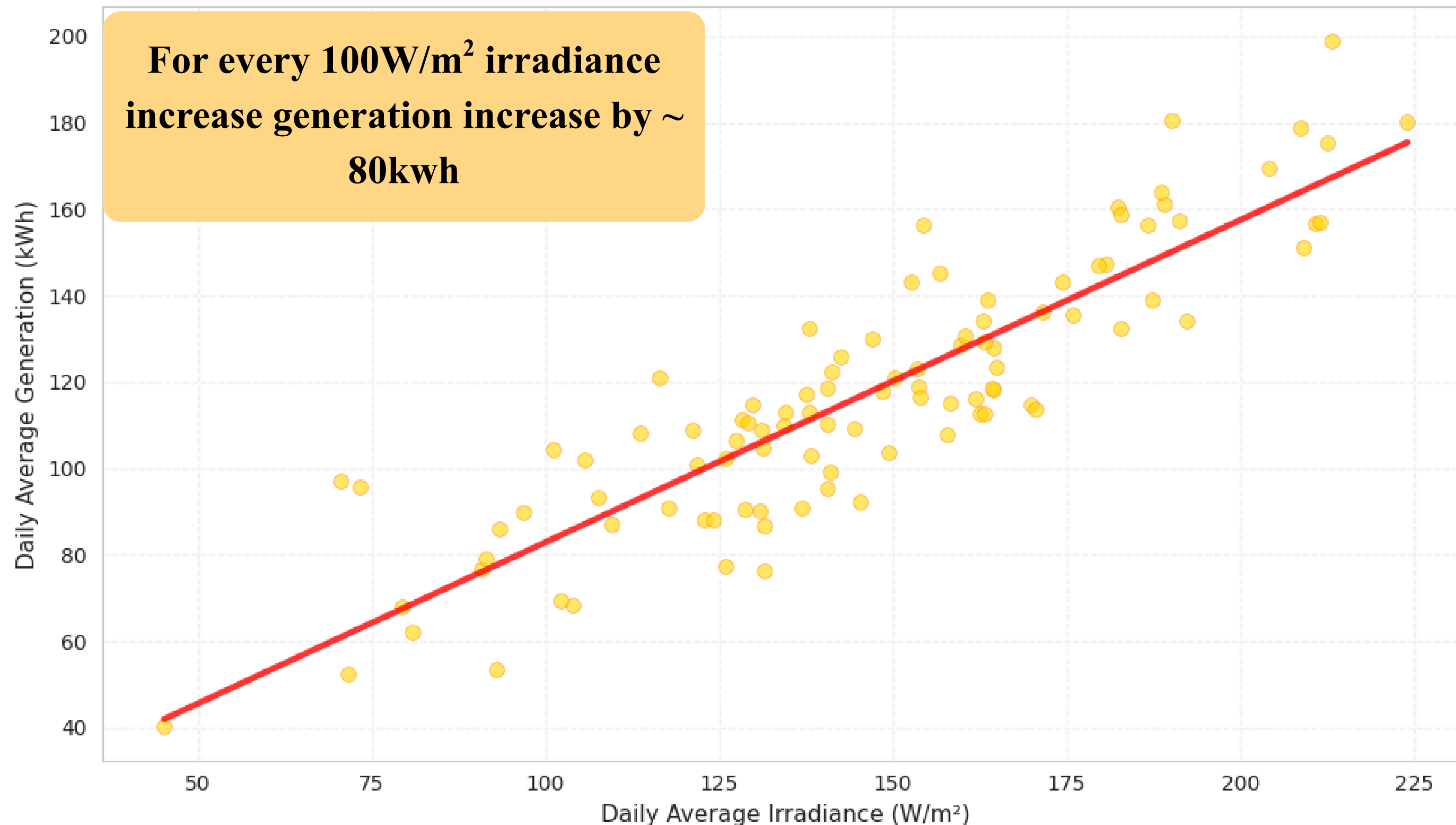
Environmental Correlation



Which Factors will affect the Power Generation?

Irradiance

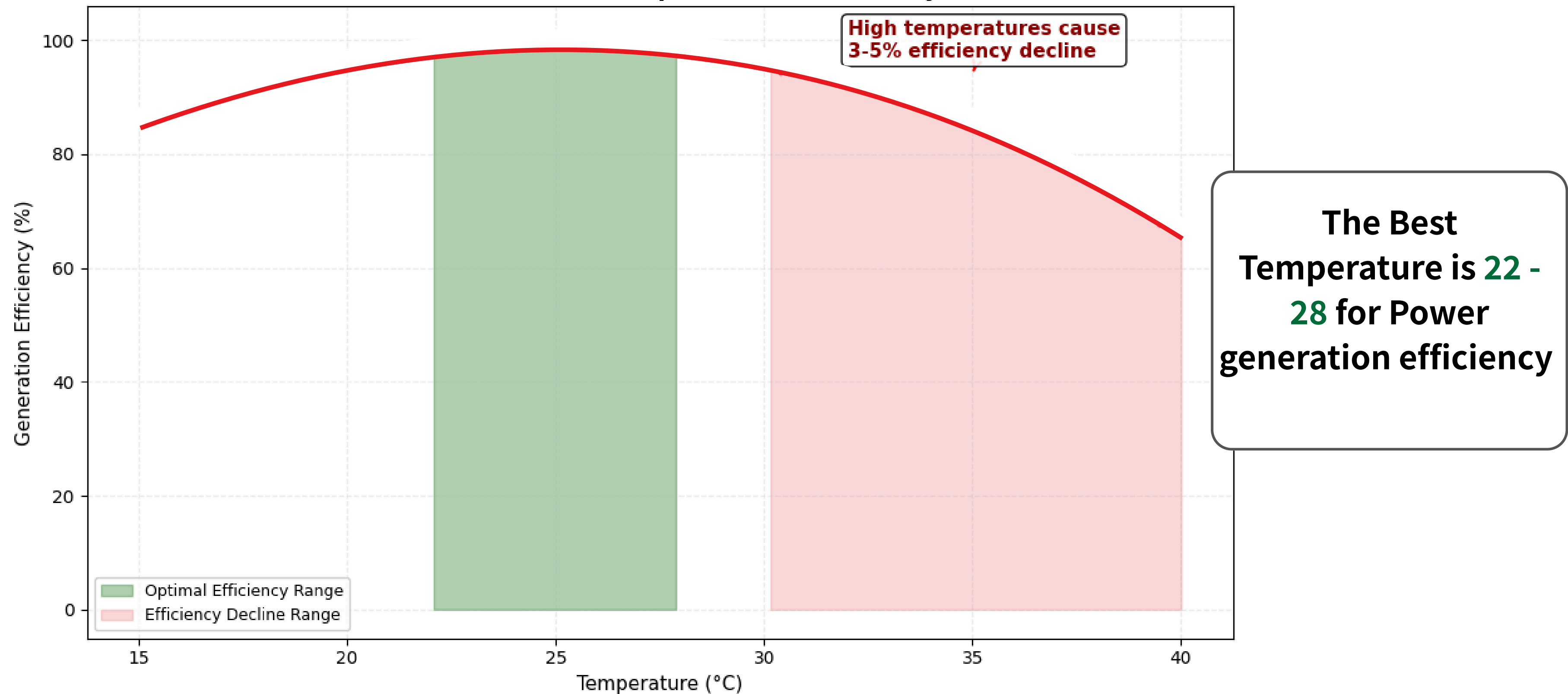
1. Irradiance Determines Generation Performance



Which Factors will affect the Power Generation?

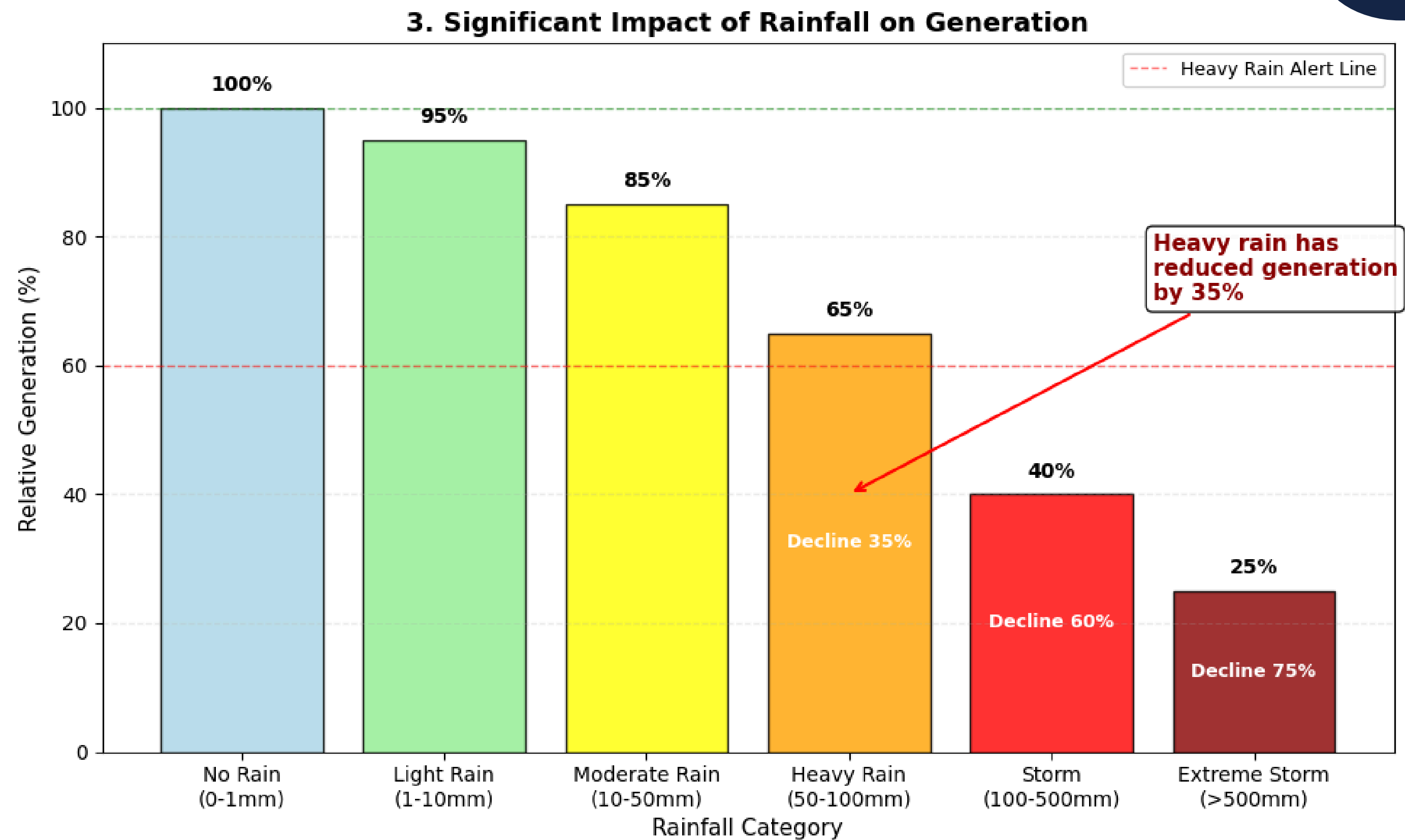
Temperature

2. Dual Effects of Temperature on Efficiency



Which Factors will affect the Power Generation?

Rainfall



How to Determine if It's Weather-Driven or Hardware-Driven?

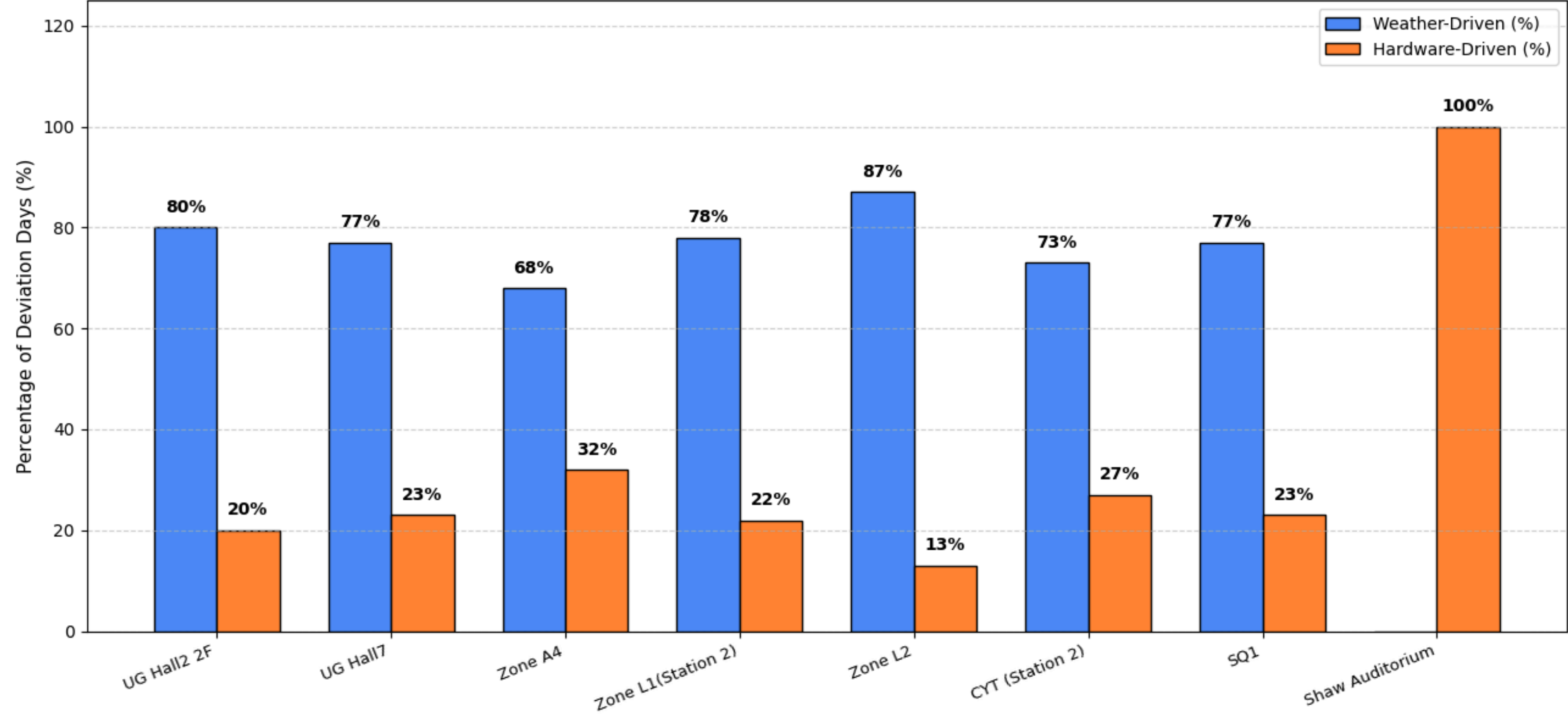
UG Hall2 2F

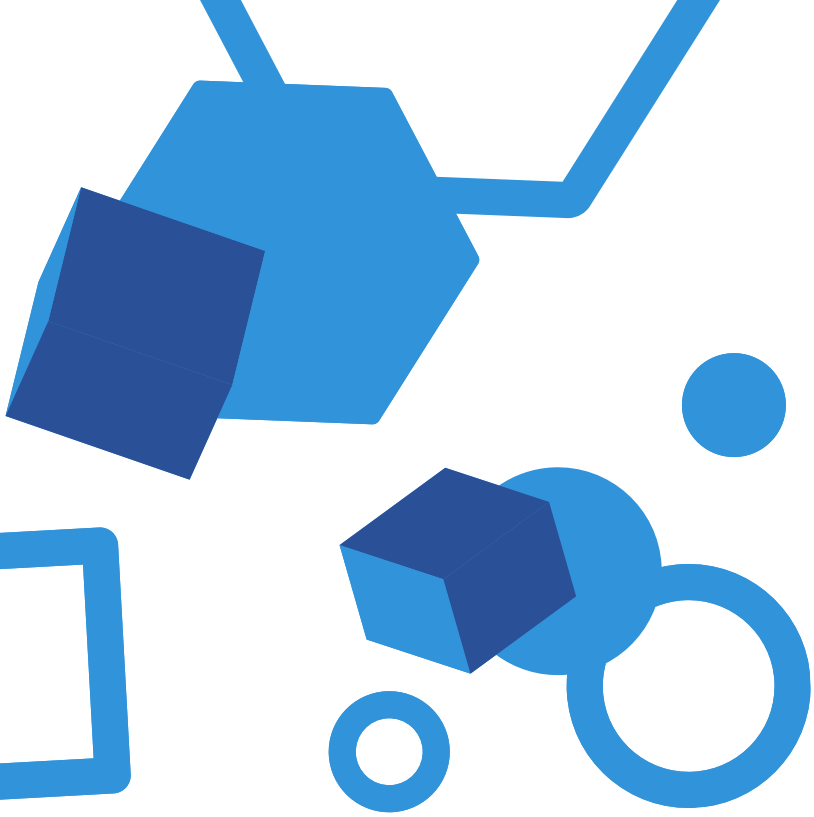
Total Down Days: 30 days

Item		Value		Explanation
Irradiance	Median (Baseline) 158.28	Max 309.38	Min 83.11	Median of daily average irradiance of the month for this location
Rainfall	Median (Baseline) 76.45	Max 515.62	Min 0.0	Median of the monthly total rainfall for this location
Weather Thresholds	Irradiance < 134.54 or Rain > 99.39			85% of irradiance median / 130% of rainfall median
Weather-Driven Days(&%)	24 days(80%)			Days that meet the weather threshold
Hardware-Driven Days(&%)	6 days(20%)			Days with normal weather but still deviated

Is the underperformance due to weather or hardware?

Weather vs Hardware Contribution
Top 8 Under-Performance Locations
(Outage/Zero days filtered out, 2021-2023)





Cross-Location Benchmarking



Statistical Outliers (Z-Score)

MOM% Performance Benchmarking

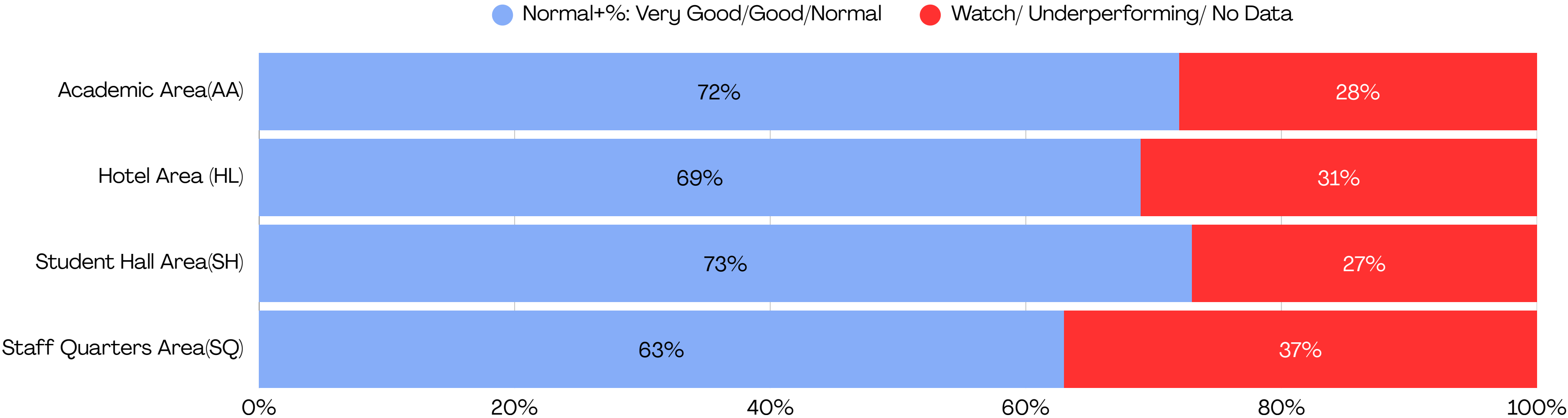
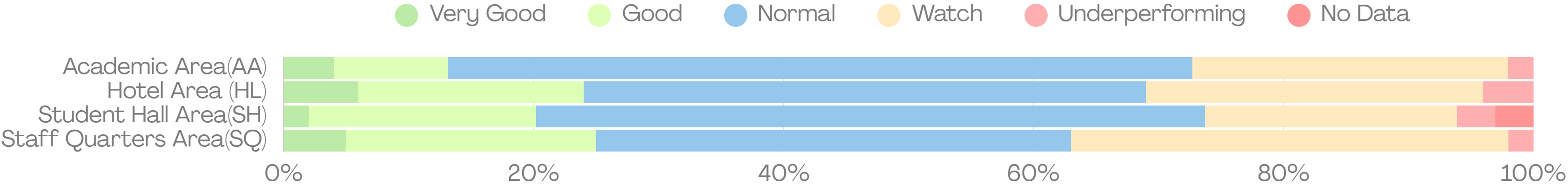
$$z = \frac{\text{Individual MOM\%} - \text{Group Average MOM\%}}{\text{Group MOM\% Standard Deviation (SD)}}$$

Level	Range (Z-Score)	Meaning
1. Very Good	$Z > 1.5$	Significantly better than the group (Top 7%)
2. Good	$0.5 < Z \leq 1.5$	Performing above average
3. Normal	$-0.5 \leq Z \leq 0.5$	Following the crowd (Most locations fall here)
4. Watch	$-1.5 < Z < -0.5$	Slightly behind; keep an eye on it
5. Underperforming	$Z \leq -1.5$	Major deviation; likely an issue/fault

Z-Score Status

MOM% Performance

• AA and SH perform well above 70%.



Z-Score: performance check

Location	Normal + %	Avg Z score	Good	Watch**	Bad**
Academic Area(AA)	72%	-0.04	0.0 + > 72%	-0.1 or < 67%	-0.1 + < 67%
Hotel Area (HL)	69%	0.28	+0.3 + > 69%	0.0 or < 64%	0.0 + < 64%
Student Hall Area(SH)	73%	-0.05	0.0 + > 73%	-0.15 or < 68%	-0.15 + < 68%
Staff Quarters Area(SQ)	63%	-0.03	0.0 + > 63%	-0.1 or < 58%	-0.1 + < 58%

**Less Avg Z-score & -5% in Normal +%

- Good:
 - Meets average level on both Normal+ % and Avg Z-score
- Watch:
 - Meets either lower Avg Z-score or Normal+ % drop >5%
- Bad:
 - Meets both lower Avg Z-score and Normal+ % drop >5%

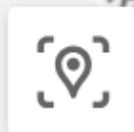
Z-score

Status

Contains

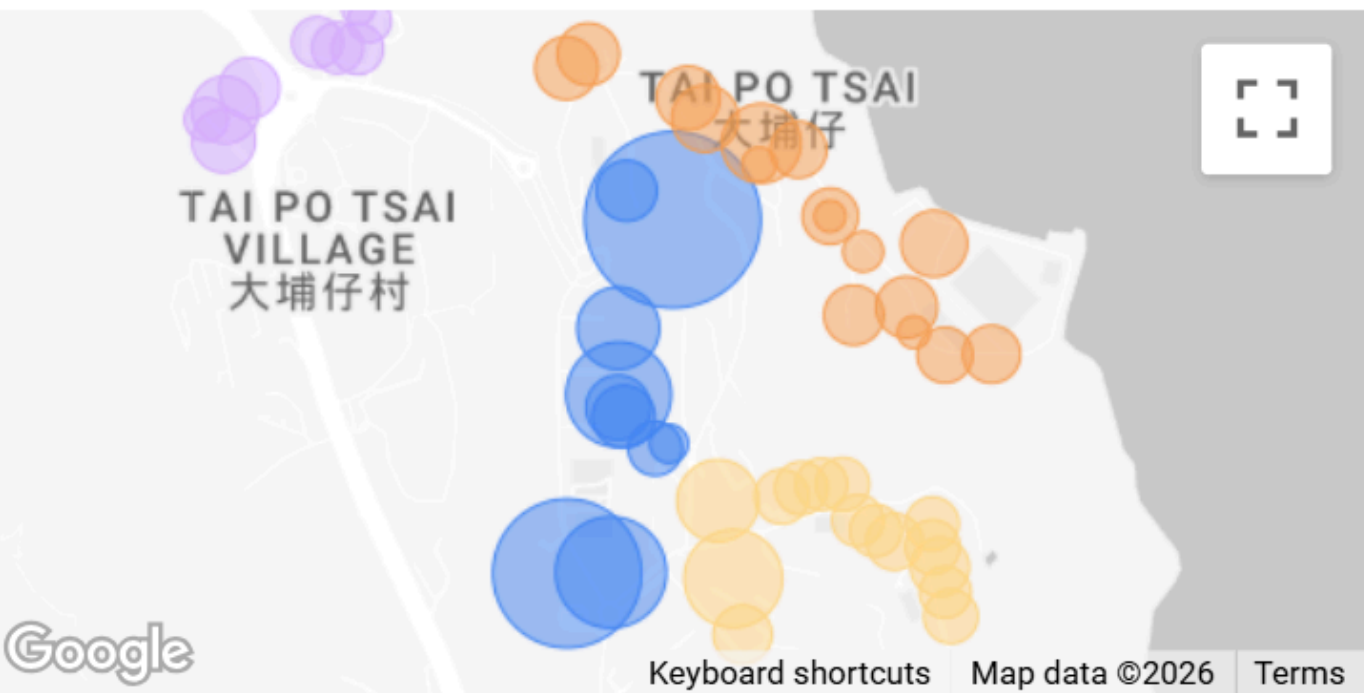


Enter a value



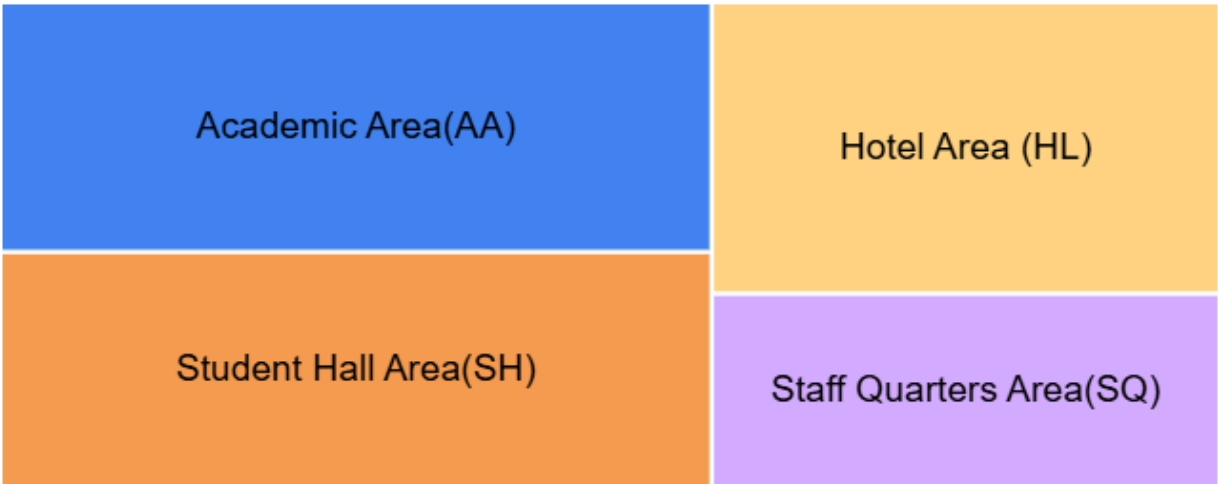
Keyboard shortcuts Map data ©2026 Terms

Status 1.Good 3. Watch 2.Normal 4. Bad



generation clean(kWh) Area

Academic Area(AA) Student Hall Area(SH) Hotel Area (HL)



location	Area	Avg Power(Kwh)
LSK NORTH	Hotel Area (HL)	6,240
Zone D	Academic Area(AA)	6,876
Zone A4	Academic Area(AA)	5,218
Zone A6	Academic Area(AA)	5,897
SQ567	Student Hall Area(SH)	4,542
Zone L1 (Station 1)	Academic Area(AA)	3,375
SQ4	Student Hall Area(SH)	3,410
Tower A	Hotel Area (HL)	4,768
Zone J2	Academic Area(AA)	3,868
SQ3	Student Hall Area(SH)	3,123
SQ1	Student Hall Area(SH)	3,043
SQ2	Student Hall Area(SH)	2,973
Zone A1	Academic Area(AA)	3,047
SQ17	Hotel Area (HL)	2,645
SQ Block R	Staff Quarters Area(SQ)	3,456
UG Hall4	Student Hall Area(SH)	2,917
Zone L1(Station 2)	Academic Area(AA)	2,638
UG Hall6	Student Hall Area(SH)	2,764

76%
in Good or Normal

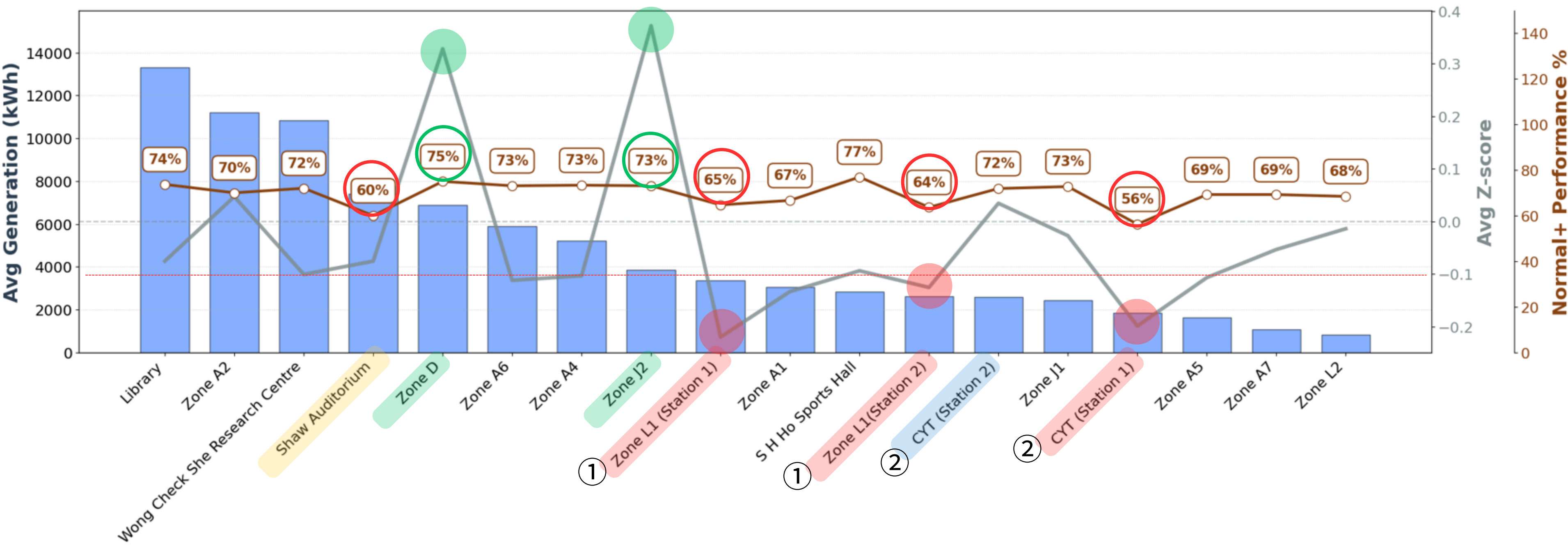
Total : 60

Good	5	8%
Normal	41	68%
Watch	7	12%
Bad	7	12%

Academic Area(AA)

Total: 18 | **Good: 2** | **Normal: 12** | **Watch: 1** | **Bad: 3**

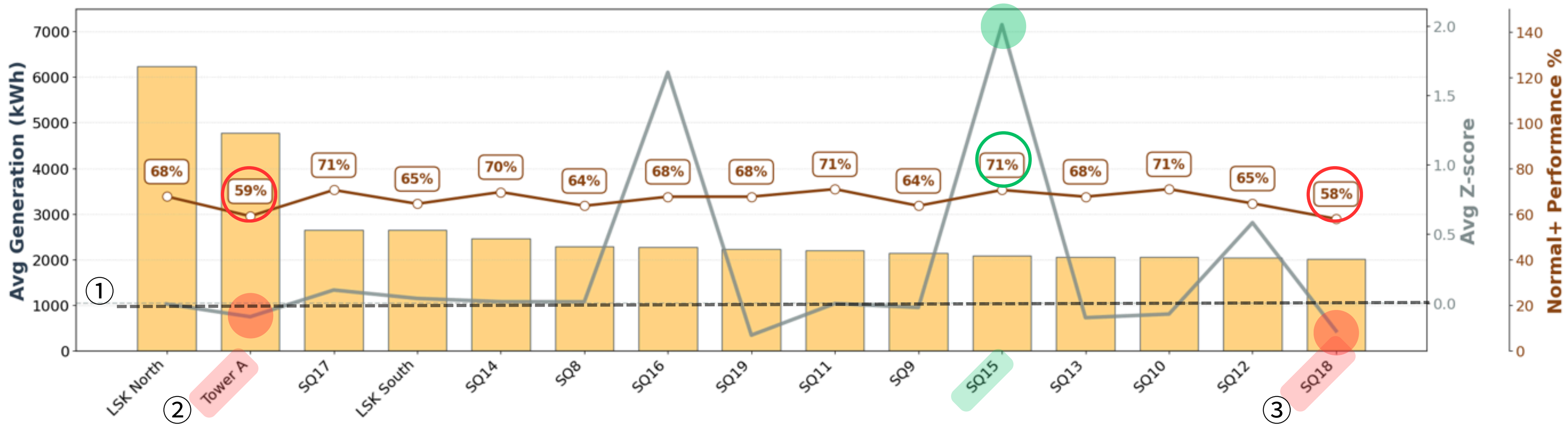
- ① Both **Zone L1 Station 1** and **Station 2** are exhibiting poor performance.
- ② In contrast, **CYT Station 1** is underperforming, while **CYT Station 2** is operating normally.



Hotel Area (HL)

Total: 15 | **Good: 1** | **Normal: 12** | **Watch: 0** | **Bad: 2**

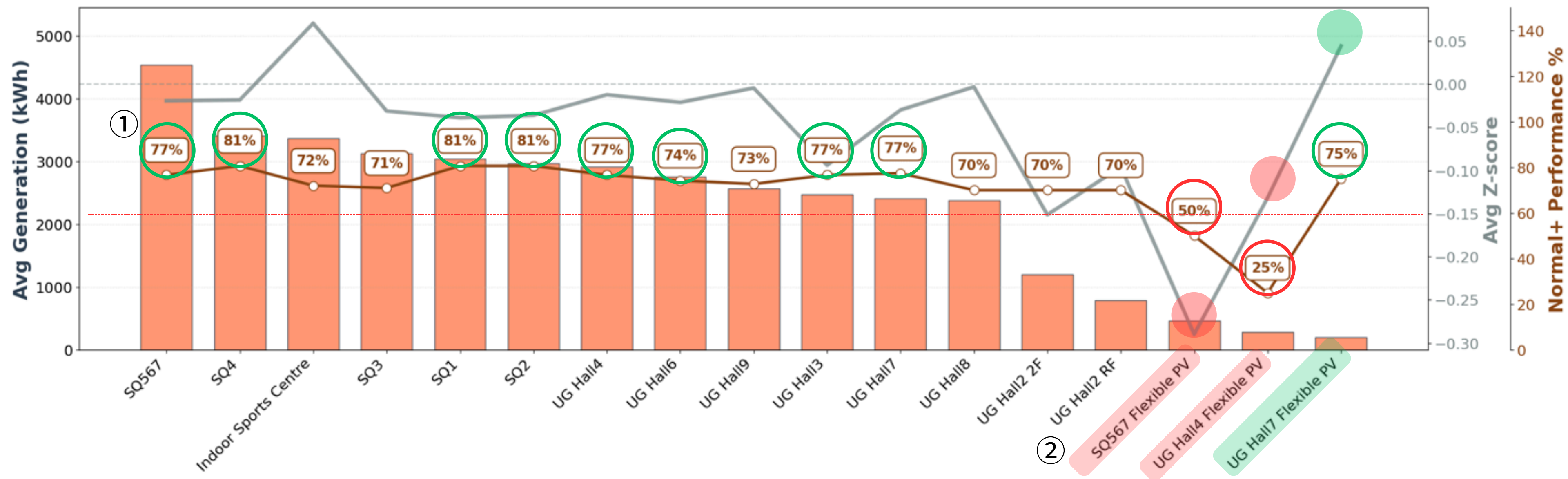
- ① HL Area: Most stable with positive Avg Z-score → Strong & consistent power output
- ② **Tower A** : 2nd highest generation in HL
- ③ Follow up with **SQ18** → Identify causes of low power



Student Hall (SH)

Total: 17 | **Good: 1** | **Normal: 14** | **Watch: 0** | **Bad: 2**

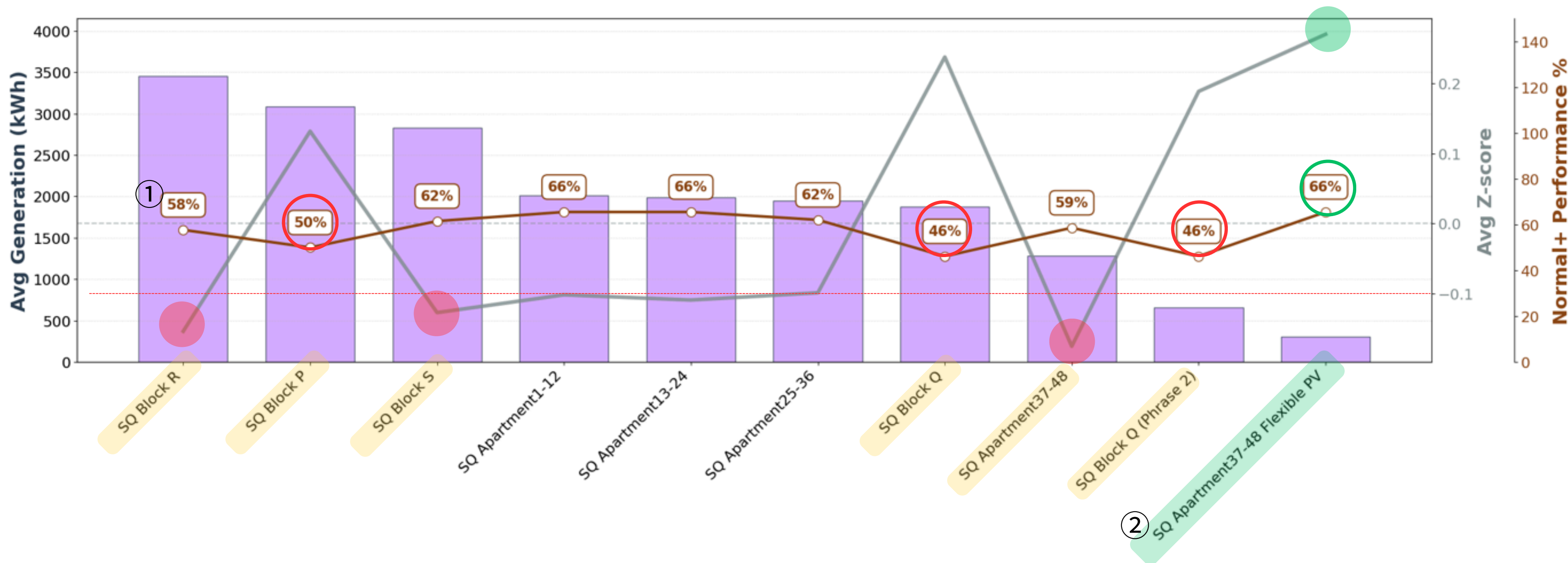
- ① SH Area: Excellent in Normal+ %→ 9 locations over 73%
- ② Flexible PV Analysis: 3 locations with Flexible PV tested
Result: **2 underperform (bad)**, **1 performs well (good)**



Staff Quarters(SQ)

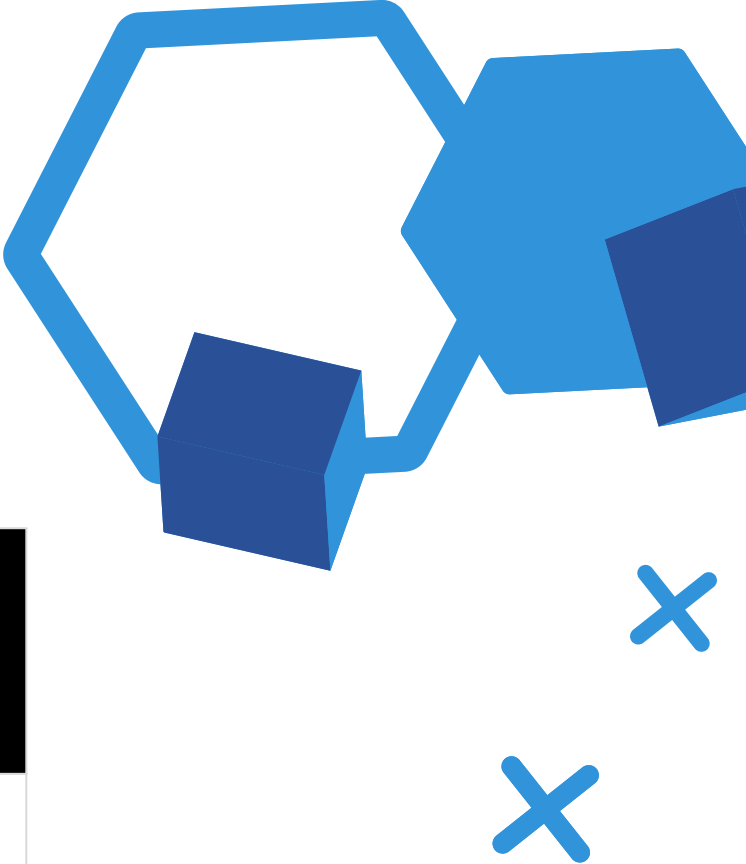
Total: 10 | **Good: 1** | **Normal: 3** | **Watch: 6** | **Bad: 0**

- ① SQ Area: Weak Performance Low Normal + % (all of them less than 66%)
- ② Flexible PV Analysis: Positive exception **SQ Apt. 37-48**



Profit & Opportunity Missing Table

Opportunity By calculating the Z-scores of underperforming data



Areas	Profit	Opportunity
Academic Area(AA)	\$8,490,729	\$117,356
Hotel Area (HL)	\$4,933,674	\$82,398
Student Hall Area(SH)	\$4,280,505	\$38,168
Staff Quarters Area(SQ)	\$1,733,001	\$2,650
Total	\$19,437,909	\$240,573

Further Action



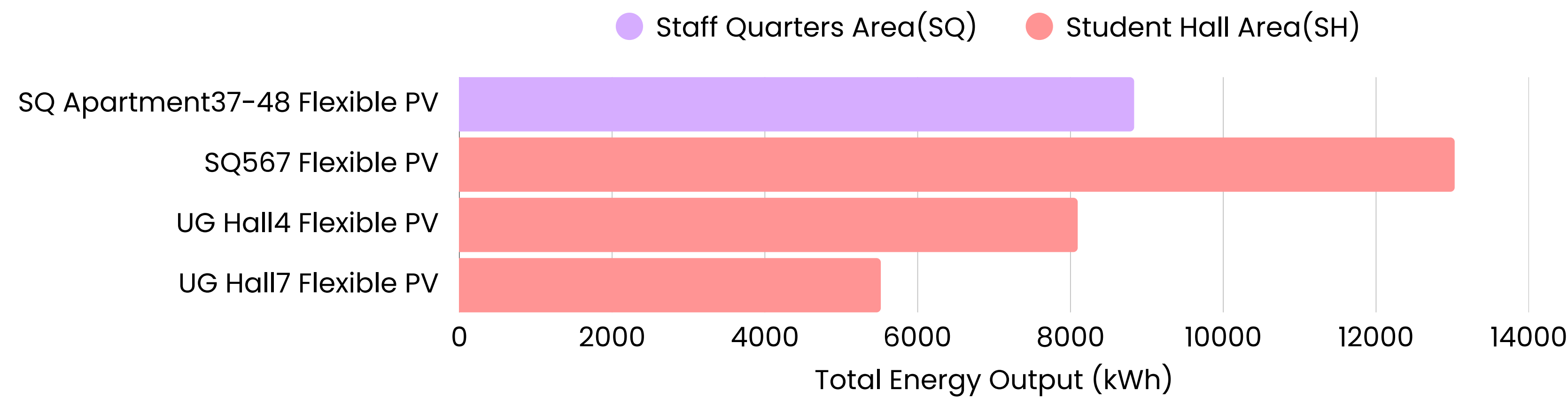
- 1)** Implement Enhanced Monitoring for High-Priority Locations
- 2)** Conduct Deep Dive into Panel-Inverter Interactions for Thin-Film Models
- 3)** Ramp Up Panel Cleaning Schedules During Winter



1) Implement Enhanced Monitoring for High-Priority Locations

Locations	Area	Z-score Status	Avg of Z-score
SQ567 Flexible PV*	Student Hall Area(SH)	Bad	-0.29
CYT (Station 1)*	Academic Area(AA)	Bad	-0.20
SQ18	Hotel Area (HL)	Bad	-0.20
Zone L1 (Station 1)	Academic Area(AA)	Bad	-0.20
UG Hall4 Flexible PV*	Student Hall Area(SH)	Bad	-0.13
Zone L1(Station 2)	Academic Area(AA)	Bad	-0.13
Tower A	Hotel Area (HL)	Bad	-0.10
Shaw Auditorium	Academic Area(AA)	Watch	-0.08

2) Conduct Deep Dive into Panel-Inverter Interactions for Thin-Film Models



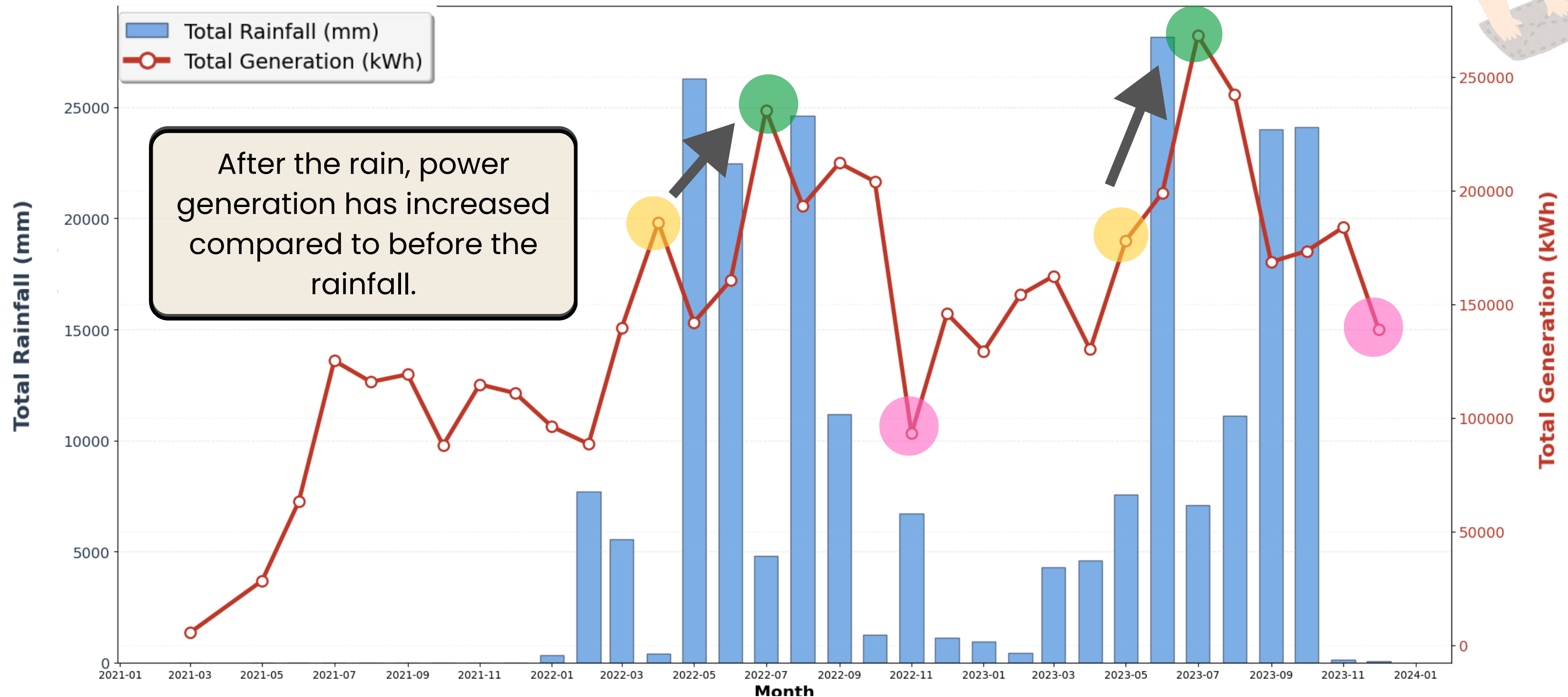
Location	Inverter Model	Zero-day Rate	Z-score
SQ Apartment37-48 Flexible PV	Sunny_Tripower_STP5_0_3AV_40	5.61	0.27
SQ567 Flexible PV	Sunny_Tripower_STP_10_0	18.36	-0.29
UG Hall4 Flexible PV	Sunny_Tripower_15000TL_30	68.84	-0.13
UG Hall7 Flexible PV	Sunny_Tripower_STP5_0_3AV_40	2.78	0.04

- System-dependent performance
- Configuration-sensitive
- Requires targeted deployment strategy

3) Ramp Up Panel Cleaning Schedules During Winter



Monthly Rainfall vs Generation Trend - All Locations Combined (2021-2023)



**Thank You
Very Much!**

